Mitosis, meiosis – what's the difference? 2a

Cells divide in order to make more cells. Usually the new cells (or daughter cells) are the same as the original ones (the parent cells). This type of cell division is called **mitosis**. Mitosis occurs when organisms grow bigger, when organisms replace lost or damaged cells, and when simple organisms reproduce asexually. Sometimes parent cells divide to develop into sex cells (or gametes); these egg and sperm contain half the normal number of chromosomes and join later during fertilisation to produce the first cell of a new offspring. This type of cell division is called **meiosis**.

When we refer to identical daughter cells, we mean that the chromosomes, which are the carriers of genetic information, are **exactly the same**. The cells themselves are smaller and similar to each other.

1 The diagram below shows a cell undergoing mitosis. Draw lines to match each description to the correct part of the diagram.

a A body cell at rest contains a cell membrane, cytoplasm and a nucleus.

b The chromosomes coil, becoming shorter and visible.

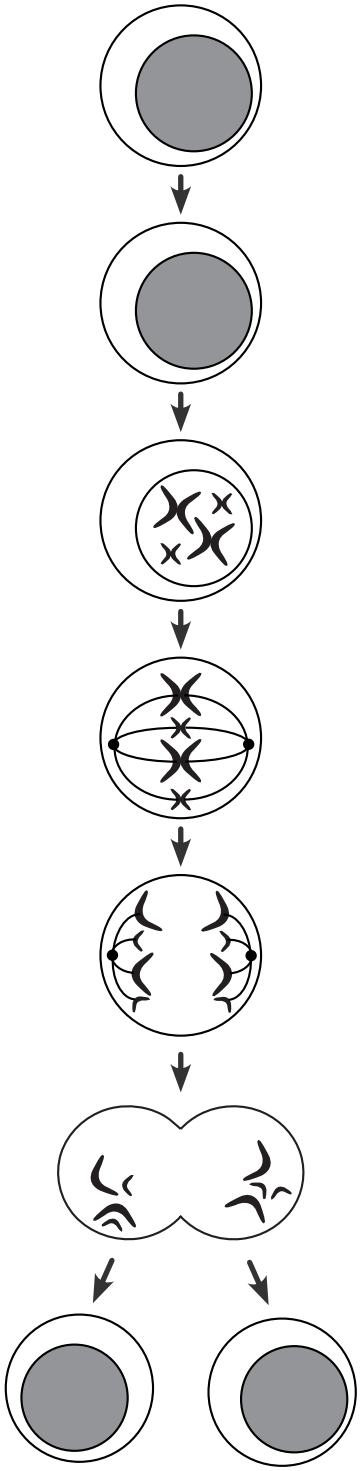
c The chromosomes congregate near the ends of the cell. The spindle dissolves and the cell starts to divide.

d The chromosomes uncoil and become invisible. Nuclear membranes form and the cell separates into two daughter cells.

e The hair-like chromosomes double.

f The nuclear membrane dissolves, the chromosomes line up along the middle of the cell and the spindle begins to grow from the ends of the cell.

g The spindle pulls on the chromosomes.
The chromosomes separate and move towards the ends of the cell.

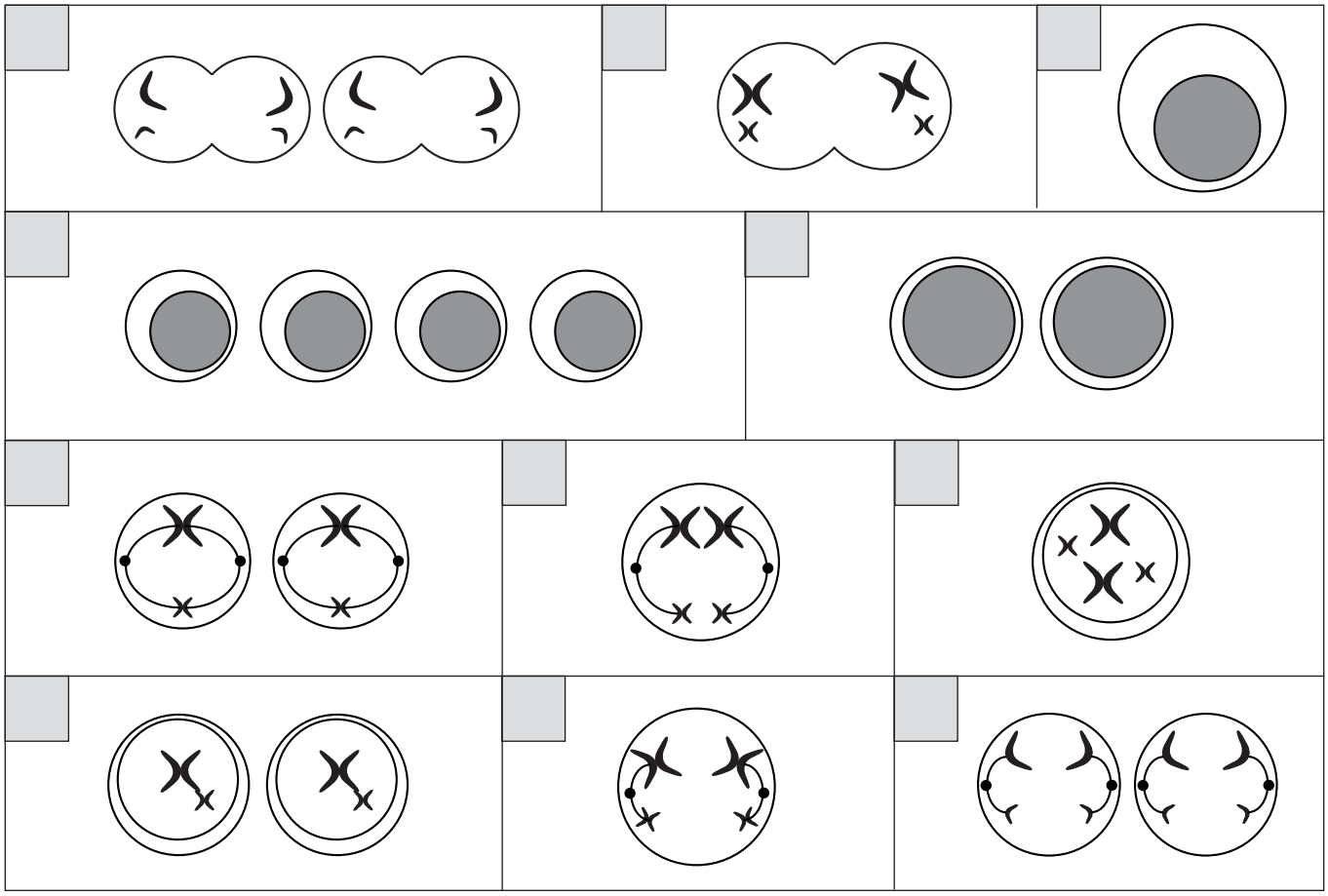


© Suada Bilali, Chris Greef and Blake Education 2010

Mitosis, meiosis – what's the difference? 2b

2 Use the information in question 1 to write in your workbook an explanation for how mitosis occurs. Remember to write a brief introduction in which you define mitosis. Then write the explanation sequence using conjunctions such as *when*, *after* and *because*. Finally, write a conclusion where you state the purpose of mitosis.

3 The diagrams below show different stages of a cell undergoing meiosis. Write the letters **a** to **k** in the grey boxes to show the sequence in which meiosis occurs.



4 In your workbook write short descriptive statements for each stage of meiosis shown above. Use the descriptive statements about mitosis on the previous pages as a model.

5 Referring to the descriptive statements you completed for question 4, write in your workbook an explanation for how meiosis occurs.

6 Complete the table below to outline the differences between meiosis and mitosis.

	Mitosis	Meiosis
Where does it occur?		
How many stages of division?		
How many cells result at the end of the division?		
In humans, how many chromosomes are there at the end of the division?		
Are the daughter cells identical to the parent cell?		
Are the daughter cells identical to each other?		

© Suada Bilali, Chris Greef and Blake Education 2010